
MORGAN R. PFAFF

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EDUCATION

University of California (UC) Berkeley and UC San Francisco

Berkeley, CA

Doctor of Philosophy in Bioengineering

October 2026

- Dissertation: “Viscoelastic, Semi-synthetic HyA Hydrogels for Soft Tissue Regeneration”
- Advisor: Kevin E. Healy, PhD
- National Science Foundation Graduate Research Fellow
- GPA 4.0/4.0
- Major area: Biomaterials Engineering & Tissue Regeneration; Minor area: Cell, Tissue, & Developmental Biology

Northeastern University

Boston, MA

Bachelor of Science in Biochemistry

May 2021

- GPA 4.0/4.0
- Honors Program; Honors in the Discipline
- Engineering & entrepreneurship study abroad program, Oxford, United Kingdom (Summer 2019)

RESEARCH EXPERIENCE

Kevin Healy Lab, UC Berkeley

Berkeley, CA

PhD Candidate

January 2022 – August 2026

- Engineered and mechanically characterized tunable, viscoelastic, semi-synthetic hyaluronic acid (HyA) hydrogels to promote muscle regeneration after rotator cuff injury and volumetric muscle loss; in a murine delayed rotator cuff repair model, the hydrogel improved recovery of muscle quality and vascularization
- Investigated hydrogel-driven mechanisms of muscle regeneration, including directed differentiation of fibro-adipogenic progenitor cells (FAPs) and associated fusion of muscle stem cells, in 3D *in vitro* cultures and cocultures
- Independently performed all single-cell RNA-seq analysis of patient-derived FAPs and used the findings to guide hydrogel design, including degradation rate, ligand presentation, drug encapsulation and release kinetics, and injectability
- Prepared hydrogels for *in vivo* injury studies with orthopedic surgery collaborators at UCSF and the University of Virginia, and analyzed and interpreted the resulting data
- Published 2 peer-reviewed papers (1 first-author), gave 5 presentations at 4 national and international conferences, and co-invented part of the hydrogel technology now in development at MuscleMatrix
- Mentored 3 PhD rotation students, 1 visiting international PhD student, 1 undergraduate, and a team of 4 Master of Translational Medicine students, and trained dozens of other students on equipment and methods

Randall Erb Lab, Northeastern University

Boston, MA

Biomaterials Undergraduate Researcher

September 2016 – July 2021

- Developed a fabrication protocol for dual-network hydrogels reinforced with magnetically responsive hydroxyapatite fibers for bone graft applications, and characterized their mechanics and microstructure using dynamic mechanical analysis and scanning electron microscopy
- Designed and analyzed COMSOL finite element simulations of hydroxyapatite fiber rotation in an alginate matrix to investigate fiber rotation as a potential new toughening mechanism
- Fabricated freeze-cast collagen conduits with aligned, double-layered porosity for peripheral nerve repair in collaboration with Dartmouth College and Harvard Medical School
- Secured independent research funding by writing a successful proposal for the competitive Project-Based Exploration for the Advancement of Knowledge (PEAK) Summit Award

Marc Hansen Lab, University of Connecticut Health Center

Farmington, CT

Summer Research Fellow

Summers 2017 & 2018

- Piloted synthetic-lethal experiments testing inhibitor and siRNA combinations in osteosarcoma cell lines and hFOB cells to identify druggable vulnerabilities in growth factor receptor signaling, using PCR and absorbance assays
- Used laser capture microdissection on histological sections to isolate osteoblast and osteoclast populations and investigate cell behavior and genetics in Paget's disease of bone
- Presented results to physicians and scientists at UConn Health; work provided preliminary data for the lab's NIH grant applications

TEACHING EXPERIENCE

University of California, Berkeley

Berkeley, CA

Reader

Fall 2026

- BIOE/MSE C118/C208: Biological Performance of Materials (Instructor: Dr. Kevin Healy; 33 students): form student design teams and grade homework, exams, written reports, and presentations
- BIOE 252: Clinical Need-Based Therapy Solutions (Instructor: Dr. Syed Hossainy; ~100 students): guide teams developing therapy concepts for unmet clinical needs, grade coursework, and coordinate guest lecturers from around the country

Guest Lecturer

2023 – 2026

- Delivered 4 full-length lectures in BIOE/MSE C118/C208 (2 on hydrogels, 1 on polymer viscoelasticity, and 1 introduction to polymers) in place of the course instructor, Dr. Kevin Healy

Graduate Student Instructor (GSI)

Fall 2022

- BIOE/MSE C118/C208: Biological Performance of Materials (Instructor: Dr. Kevin Healy; ~40 students)
- Led a weekly discussion section reviewing lecture material and held weekly office hours
- Administered and graded homework and exams, and evaluated ~10 team design projects, including 15-page reports proposing biomaterial solutions to unmet clinical needs and a series of pitch, 10-minute, and 20-minute presentations

TRANSLATIONAL, REGULATORY, & INDUSTRY EXPERIENCE

MuscleMatrix

Berkeley, CA

Scientific Consultant

August 2026 – Present

- Consult for an early-stage startup spun out in part from doctoral research that is developing an injectable, cell-free hyaluronic acid scaffold to support muscle regeneration after rotator cuff injury
- Help build company operations from the ground up, including regulatory compliance, environmental health and safety (EHS), laboratory buildout and equipment procurement, and manufacturer and supplier sourcing
- Contribute to FDA regulatory strategy for approval as a medical device or combination product, and write grant applications
- Conduct formulation development, characterizing hydrogel injectability and mechanics, and how drug loading affects them, to inform a planned drug-eluting version
- Train new scientific staff on the hydrogel platform, complex technical procedures, and laboratory operations

Vertex Pharmaceuticals

Boston, MA

Exploratory Research Co-op

January 2020 – June 2020

- Developed and optimized a reporter assay to measure transcription factor activity in potentially druggable pathways for sickle cell disease (SCD), giving the team a faster, more accurate readout of compound effectiveness
- Generated stable cell lines by molecular cloning of recombinant vectors for genomic integration
- Independently planned, executed, and analyzed experiments, and presented results and journal club articles to the SCD exploratory research team

- Screened sgRNAs to identify the most effective candidates for the company's pipeline diseases, introducing CRISPR-mediated edits via nucleofection, transfection, and viral transduction
- Programmed and optimized high-throughput screening assays on automated liquid handlers, including a comparative performance evaluation of the Biomek i7 and its predecessor, the Biomek FX, enabling more complex, higher-throughput sgRNA screening
- Analyzed data in Prism, Tableau, and FlowJo and presented findings to the screening group and other departments

PUBLICATIONS & PRESENTATIONS

Publications

1. Wang, V. Y., Zhu, Y., Spaulding, K. A., **Pfaff, M. R.**, Neiman, G., Takaba, K., Chen, N., Parekh, K., Hasaballa, A., Haraldsson, H., Baker, A., Saloner, D., Wallace, A. W., Ziats, N. P., Healy, K. E., Ratcliffe, M. B. NIPAAm-co-PEG4000 thermoresponsive hydrogel with ROS scavenging properties improves in vivo border zone contractility and reduces myocardial remodeling in sheep after MI. *Acta Biomaterialia*, 2026, 211, 48–60. doi.org/10.1016/j.actbio.2025.06.046
2. **Pfaff, M. R.**, Wague, A., Davies, M., Killaars, A. R., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Viscoelastic HyA hydrogel promotes recovery of muscle quality and vascularization in a murine model of delayed rotator cuff repair. *Advanced Healthcare Materials*, 2025, 14, 2403962. doi.org/10.1002/adhm.202403962

Oral Presentations

1. **Pfaff, M. R.**, Wague, A., Davies, M., Killaars, A. R., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Viscoelastic HyA hydrogel promotes recovery of muscle quality and vascularization in a murine model of delayed rotator cuff repair. Presented at: Biological Scaffolds for Regenerative Medicine (BSRM); May 2025; Napa, CA.
2. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Presented at: Biomedical Engineering Society (BMES) Annual Meeting; October 11–14, 2023; Seattle, WA.
3. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Presented at: Society for Biomaterials (SFB) Annual Meeting & Exposition, “Biomaterials for Regenerative Engineering”; April 21, 2023; San Diego, CA.
4. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Presented at: Materials Research Society 2020 Spring/Fall Meeting and Exhibit; November 28, 2020; Virtual.
5. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Presented at: Northeastern Regional Student Conference of the Society for Experimental Mechanics; June 22, 2020; Dartmouth, MA.

Poster Presentations

1. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Presented at: Orthopaedic Research Society (ORS) 50th International Musculoskeletal Biology Workshop; July 22–27, 2023; Midway, UT.
2. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Presented at: Society for Biomaterials (SFB) Annual Meeting & Exposition; April 2023; San Diego, CA.
3. **Pfaff, M. R.**, Erb, R. M. Understanding rod rotation in fiber reinforced hydrogels: a potential toughening mechanism. Presented at: Research, Innovation, and Scholarship Expo; April 15, 2021; Virtual.
4. **Pfaff, M. R.**, Alander, C. B., Hansen, M. F. Synthetic lethal approaches to treating osteosarcoma. Presented at: University of Connecticut Health Center Summer Research Fellowship Program; July 27, 2017; Farmington, CT.

5. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Presented at: Research, Innovation, and Scholarship Expo; April 13, 2017; Boston, MA.
6. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Presented at: National Conference on Undergraduate Research; April 6–8, 2017; Memphis, TN.

FELLOWSHIPS & AWARDS

National Science Foundation Graduate Research Fellowship (\$138,000)	2021
Research, Innovation, and Scholarship Expo (RISE) Excellence in Research Award (\$2,000)	2021
PEAK Summit Award, undergraduate research grant (\$3,000)	2020
PEAK “Shout It Out!” Award, conference travel grant	2020
RISE Innovation Alley Selection (poster selected for highest impact and commercial potential)	2017

LEADERSHIP & SERVICE

Food Not Bombs East Bay **September 2024 – Present**

- Lead a weekly shift, picking up meals prepared by volunteer cooks and serving 25–40 free vegetarian meals to unhoused and low-income community members
- Transformed an unstaffed curbside drop-off into an in-person meal service, creating a more personal experience and building ongoing relationships with community members

Sigma Xi, The Scientific Research Honor Society (Associate Member) **March 2020 – Present**

Vice President, UC Berkeley Chapter 2021 – Present

- Independently planned and executed the Sigma Xi–ATP Bio Research Symposium & Poster Session, open to students at all levels, handling everything from securing a funding partner to recruiting presenters and day-of logistics; drew ~25 presenters, a keynote speaker, and ~150 attendees, who voted on a best-poster prize
- Co-organize the chapter’s semi-annual Grants in Aid of Research (GIAR) program, which awards research grants of up to \$750 to undergraduate, master’s, and PhD students across STEM
- Founded and chair the Committee for Student Advancement (CSA), growing it to ~40 students, including preliminary members not yet eligible for associate membership
- Host networking events and student and guest talks, including a talk by the Editor-in-Chief of *American Scientist*
- Lead chapter recruitment through outreach to campus organizations and departments, and manage chapter operations, including scheduling and records

Diversity & Inclusion Chair, Northeastern University Chapter 2020 – 2021

- Expanded outreach to first-generation, culture-specific, and other underrepresented student organizations, and hosted a campus-wide diversity, equity, and inclusion panel

BioEngineering High School Competition (BioEHSC) **April 2022 – April 2026**

- Academic Judge: evaluated 10–15 high school team pitches each spring in a full-day bioscience pitch competition, assessing depth of knowledge through Q&A and providing feedback

Northeastern University Club Sports **September 2016 – June 2021**

- Triathlon Team: President/Captain (2019–2020), Chief Financial Officer (2018–2019), and Chief Procurement Officer (2017–2018)
- Club Sports Council (2019–2021): served as liaison between teams and Club Sports administration

Adventure Challenge Volunteer **September 2018 – December 2019**

- Planned and led weekly swimming and pool games with a Boston Children’s Hospital patient with a disability to foster movement, practice social skills, and build connection

Northeastern University Resident Assistant**August 2017 – July 2019**

- Served as a live-in mentor for ~45 first-year women each year from around the world pursuing careers in STEM, helping them set and achieve goals, connect with scientists and professors, and navigate issues including academic performance, inclusion, and substance use
- Designed and implemented educational, recreational, and networking programs, including Q&A panels with graduate students and professors, leadership and team-building workshops, and faculty mentorship events

Advancement Via Individual Determination (AVID) Tutor**September 2018 – December 2018**

- Led a weekly group of 6–8 students at Fenway High School in the AVID college-readiness program, teaching collaborative problem-solving and critical thinking for math and science

CLINICAL EXPERIENCE**Tufts Medical Center, Oncology and Hematology****Boston, MA**

Clinical Care Technician

January 2019 – December 2019

- Provided direct care for 6–10 oncology and hematology patients per shift, including support with mobility, hygiene, and meals
- Monitored vital signs and blood glucose, escalated abnormal findings to nursing staff, and supported clinical teams during examinations, procedures, and emergencies

Fallon Ambulance Service**Quincy, MA**

Emergency Medical Technician (EMT-Basic)

September 2018 –

April 2019

- Assessed, treated, and transported ill and injured patients, providing immediate medical and psychological care en route to hospitals and specialty facilities
- Documented vitals, symptoms, medical history, allergies, and medications per company protocols while maintaining a calm, safe environment for patients and families

Dental Office of Dr. Andrew J. Ponichtera, DMD, MS**Simsbury, CT**

Dental Assistant

June 2016 – August 2016

- Supported chairside procedures including fillings, root canals, and placement of crowns, implants, bridges, and dentures
- Managed front-desk operations and supply inventory, including patient scheduling, records, and ordering

SKILLS**Biomaterials Synthesis & Mechanical Testing**

- Hydrogel synthesis and chemical modification, drug loading and release, rheology, injectability testing, mechanical testing (dynamic mechanical analysis, tensile, burst, and lap shear), freeze casting, lyophilization

Materials Characterization & Microfabrication

- NMR spectroscopy, X-ray photoelectron spectroscopy (XPS), scanning electron microscopy (SEM), microfabrication and photolithography

Cell & Molecular Biology

- Primary and 3D cell culture, immunocytochemistry, confocal and fluorescence microscopy, histology, flow cytometry, qPCR/ddPCR, Western blot, molecular cloning, CRISPR-Cas9 gene editing (transfection, nucleofection, viral transduction), automated liquid handling

Data Analysis & Computation

- R and Python (including single-cell RNA-seq analysis), GraphPad Prism, MATLAB, COMSOL Multiphysics (finite element analysis)

Regulatory & Translational

- FDA regulatory pathways and product classification, environmental health and safety (EHS) compliance, laboratory buildout and procurement, grant writing